

LUPIN, NU, Canada

WMO: 714700

Lat: 65.760N

Lon: 111.250W

Elev: 488

StdP: 95.60

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.4	-37.8	-43.7	0.1	-38.9	-41.8	0.1	-37.4	15.2	-22.3	13.8	-24.7	3.0	270	1.001

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	22.4	14.0	20.4	13.3	18.2	12.1	15.0	20.6	13.9	19.0	12.7	17.4	5.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.7	9.7	16.8	11.5	8.9	15.8	10.3	8.2	14.9	43.8	20.5	40.6	19.0	37.4	17.4	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.8	10.3	DB	-41.2	25.9	1.9	1.7	-42.6	27.1	-43.7	28.1	-44.7	29.0	-46.1	30.3	(4)
(5)			WB	-40.9	16.7	1.7	1.0	-42.1	17.5	-43.1	18.1	-44.1	18.6	-45.4	19.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.7	-28.4	-26.2	-22.8	-15.8	-4.8	7.2	11.9	9.3	2.8	-7.2	-18.5	-25.5
	DBStd	15.39	6.38	6.05	7.02	6.65	6.05	4.94	3.91	3.93	4.52	5.38	6.08	6.21
	HDD10.0	7359	1190	1012	1017	774	458	114	26	61	217	535	854	1101
	HDD18.3	10248	1448	1246	1276	1024	716	335	201	282	464	793	1104	1359
	CDD10.0	154	0	0	0	0	0	29	84	38	3	0	0	0
	CDD18.3	2	0	0	0	0	0	0	2	1	0	0	0	0
	CDH23.3	27	0	0	0	0	0	7	14	6	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	0	1	0	0	0	0
Wind	WSAvg	4.8	4.5	4.6	4.9	4.7	4.9	4.8	4.6	4.9	5.5	5.3	4.6	4.0
Precipitation	PrecAvg	293	7	7	14	14	17	26	36	63	42	29	17	14
	PrecMax	421	17	19	39	29	40	64	86	153	83	52	31	31
	PrecMin	224	2	0	4	2	2	0	11	26	7	12	7	3
	PrecStd	50	4	5	9	8	10	18	21	30	20	11	7	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-9.8	-9.9	-4.7	1.9	12.7	24.2	25.2	23.8	16.6	5.4	-2.4	-7.9
		MCWB	-10.2	-10.2	-5.2	0.2	7.5	13.9	15.5	14.1	11.1	4.1	-2.8	-8.4
	2%	DB	-12.9	-12.4	-7.1	-0.3	8.2	20.9	23.1	21.0	13.5	2.1	-5.5	-10.1
		MCWB	-13.2	-12.8	-7.6	-1.1	4.8	12.9	14.3	14.0	9.6	1.6	-6.0	-10.4
	5%	DB	-16.0	-14.7	-9.3	-3.0	5.2	18.3	21.2	18.1	11.0	0.7	-7.4	-13.1
		MCWB	-16.4	-15.0	-9.8	-3.6	2.8	11.6	13.7	12.4	8.6	0.2	-7.7	-13.5
	10%	DB	-18.6	-17.2	-12.1	-5.5	3.0	15.8	19.4	15.6	9.2	-0.4	-9.7	-16.1
		MCWB	-19.0	-17.6	-12.6	-6.1	1.3	10.4	12.9	11.3	7.3	-0.8	-10.1	-16.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-10.2	-10.2	-5.2	0.4	7.9	15.2	17.1	15.8	12.0	3.9	-2.7	-8.3
		MCDB	-9.8	-10.0	-4.8	1.6	12.7	22.5	22.4	21.0	15.1	4.5	-2.4	-8.1
	2%	WB	-13.3	-12.7	-7.6	-1.1	4.8	13.5	15.6	14.2	10.0	1.6	-5.9	-10.5
		MCDB	-13.0	-12.4	-7.2	-0.3	7.9	19.7	21.3	19.2	12.7	2.0	-5.6	-10.2
	5%	WB	-16.3	-15.0	-9.7	-3.6	2.9	12.1	14.5	13.1	8.7	0.3	-7.8	-13.5
		MCDB	-16.1	-14.8	-9.3	-3.0	5.1	17.8	19.6	17.4	11.0	0.6	-7.4	-13.1
	10%	WB	-18.9	-17.5	-12.5	-6.0	1.4	10.6	13.4	11.8	7.4	-0.9	-10.1	-16.4
		MCDB	-18.6	-17.2	-12.1	-5.5	2.9	15.4	18.2	15.0	9.0	-0.5	-9.7	-16.0
Mean Daily Temperature Range	5% DB	MDBR	6.5	6.7	7.8	8.6	7.3	8.9	9.3	7.6	5.5	4.7	6.3	6.4
		MCDBR	8.3	8.2	9.0	8.8	9.2	13.1	12.1	11.4	9.0	4.5	7.2	9.0
		MCWBR	8.0	8.2	8.9	8.2	6.5	6.9	5.7	5.6	4.1	7.2	9.0	
	5% WB	MCDBR	8.2	8.1	8.9	8.8	8.7	12.4	11.2	10.4	8.3	4.3	7.2	9.0
		MCWBR	8.0	8.0	8.8	8.3	6.3	6.8	6.3	5.7	5.5	4.0	7.2	9.0
Clear-Sky Solar Irradiance	taub	0.147	0.222	0.242	0.283	0.311	0.326	0.343	0.309	0.283	0.243	0.153	0.140	
	taud	1.915	2.147	2.262	2.194	2.263	2.455	2.435	2.537	2.549	2.233	1.915	1.935	
	Ebn at Noon	371	679	852	885	890	887	857	862	811	636	324	96	
	Edh at Noon	18	52	77	107	112	96	95	77	60	47	17	4	
All-Sky Solar Radiation	RadAvg	0.11	0.77	2.51	4.92	6.30	5.63	5.27	3.66	1.96	0.57	0.16	0.02	
	RadStd	0.01	0.04	0.13	0.21	0.46	0.40	0.35	0.33	0.26	0.09	0.03	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page